A COMPUTING DEVICE WITH HARDWARE-ENFORCED GRID-ISOLATED OPERATION

Reinaldo Javier Aguiar Marcano

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Title: A COMPUTING DEVICE WITH HARDWARE-ENFORCED GRID-ISOLATED OPERATION

Inventor: Reinaldo Javier Aguiar Marcano, a citizen of the United States, residing at 2302 Britton Ridge Drive, Katy, TX 77494.

Abstract

A secure computing device is disclosed which can only achieve a fully operational state when it is physically disconnected from an external electrical power grid. The device contains an internal rechargeable power source. A hardware-based switching mechanism continuously monitors whether the internal power source is being charged by an external source. If it detects that external power is being drawn, the switching mechanism physically interrupts the power supply to all non-essential, high-emission peripherals, including but not limited to video outputs, USB buses, and all wireless and wired network interfaces. Only when the device is disconnected from the external power grid does the switching mechanism permit power to flow to these peripherals, allowing the device to boot into a fully functional state. This hardware-level constraint prevents sophisticated adversaries from remotely compromising the device through software and ensures that the device can only be used in an electrically isolated, "off-grid" state, thereby frustrating surveillance techniques that rely on monitoring power grid consumption to time remote attacks.

Background of the Invention

Sophisticated adversaries, including state-sponsored actors, are known to monitor the power consumption of a target's residence or facility. Fluctuations in power draw can indicate when a target is actively using a computer. For targets using RF-based communication systems such as cellular modems or satellite terminals, this information is highly valuable. The adversary can use the detection of computer activity (via power monitoring) to optimally time the deployment of a Man-in-the-Middle (MITM) attacker or other forms of RF-based surveillance and attack.

Software-based solutions that attempt to limit a computer's functionality are vulnerable to being disabled by malware. A sufficiently advanced persistent threat can gain control over the operating system and bypass any software-level restrictions. Therefore, there is a need for a computing device that enforces a secure, off-grid operational mode at the hardware level, in a way that cannot be overridden by software compromise.

Summary of the Invention

The present invention is a computing device designed to counter power-monitoring-based surveillance by enforcing a physically-isolated operational mode. The device's core principle is that it will not power its primary I/O and communication peripherals while it is connected to mains electricity.

The device comprises a main computing module (CPU, memory, storage), an internal rechargeable battery, a charging circuit for the battery from an external power source, and a novel hardware switching mechanism. This switching mechanism, preferably implemented with physical relays, is the core of the invention. It senses the flow of current from the external power source to the charging circuit. If current is flowing, meaning the device is plugged in and charging, the relays are held in an open state, physically severing the power connections to peripherals like the display controller, USB ports, Wi-Fi card, and Ethernet port. In this "charging" state, only the core components needed to manage the battery can be active.

Only when the device is physically disconnected from the external power source does the sensing circuit detect a cessation of charging current. This triggers the relays to close, completing the circuits and supplying power from the internal battery to all the peripheral

controllers. This allows the full operating system to boot and the device to become fully functional. This hardware-interlocked design ensures that the device can only be used for sensitive tasks when it is electrically isolated, making it impossible for an adversary to correlate its usage with fluctuations in the power grid.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a block diagram of the secure computing device's architecture, showing the hardware switching mechanism and its control over peripheral power.
FIG. 2 is a state diagram illustrating the two mutually exclusive operational states of the device: the "Charging / Peripherals Disabled" state and the "Off-Grid / Peripherals Enabled" state.

Detailed Description of the Invention

Referring to **FIG. 1**, the architecture of the secure computing device (100) is shown. The device has an external power input (101) that feeds a charging circuit (102). The charging circuit recharges the internal battery (103). The core of the invention is the current sensing and switching mechanism (104). This mechanism includes a current sensor (105) that monitors the flow of electricity from the external input to the charger.

When the device is plugged in, the sensor (105) detects an active charging current. This detection signal controls a set of physical power relays (106). As long as a charging current is detected, the relays (106) are held open, interrupting the power flow from the battery (103) to the non-essential peripheral bus (107). This bus powers components such as the network interfaces (Wi-Fi, Ethernet) (108), USB ports (109), and video outputs (110). In this state,

the main computing core (111) may be in a low-power standby mode, but it cannot fully operate.

When the external power input (101) is disconnected, the sensor (105) detects the cessation of charging current. This causes the switching mechanism (104) to close the relays (106), allowing the battery (103) to supply power to the peripheral bus (107), thus enabling all peripherals and allowing the main computing core (111) to boot into a fully operational state.

FIG. 2 illustrates the two operational states. When the device is connected to the grid (201), it enters the "Charging State" (202). In this state, peripherals are powered off (203). When the device is disconnected from the grid (204), it transitions to the "Off-Grid State" (205). In this state, peripherals are powered on (206) and the device is fully functional. The transition from the Off-Grid State back to the Charging State is triggered by reconnecting to the grid (207), which immediately cuts power to the peripherals. This hardware-enforced logic ensures the device cannot be fully used while being monitored via the power grid.

Claims

What is claimed is:

1. A computing apparatus, comprising:
    - a central processing unit;
    - a rechargeable internal power source for powering the central processing unit;
    - a power input for receiving external electrical power to recharge the internal power source;
    - a plurality of peripheral devices, including at least one data communication interface; and
    - a hardware-based switching mechanism communicatively coupled to the power input and the plurality of peripheral devices, wherein the switching mechanism is configured to physically prevent the supply of power from the internal power source to the plurality of peripheral devices when it detects that the internal power source is being recharged by external electrical power.
2. The apparatus of claim 1, wherein the switching mechanism comprises one or more physical relays.

3. The apparatus of claim 1, wherein the switching mechanism includes a current sensor to detect the flow of external electrical power to the internal power source.

4. The apparatus of claim 1, wherein the plurality of peripheral devices includes a video output controller and a USB bus controller.

5. The apparatus of claim 1, wherein the data communication interface is a wireless network interface or a wired network interface.

6. A method for enhancing the security of a computing device, comprising the steps of:

    - detecting, using a hardware sensor, whether the device's internal battery is being charged by an external power source;
    - in response to detecting that the battery is being charged, physically interrupting, via a hardware switch, the flow of power to one or more peripheral components of the device, including all network communication interfaces; and
    - in response to detecting that the battery is not being charged, allowing power to flow to the one or more peripheral components.

7. The method of claim 6, wherein the hardware switch is a relay.

8. The method of claim 6, wherein the peripheral components whose power is interrupted also include video display controllers and data port controllers.

Drawings

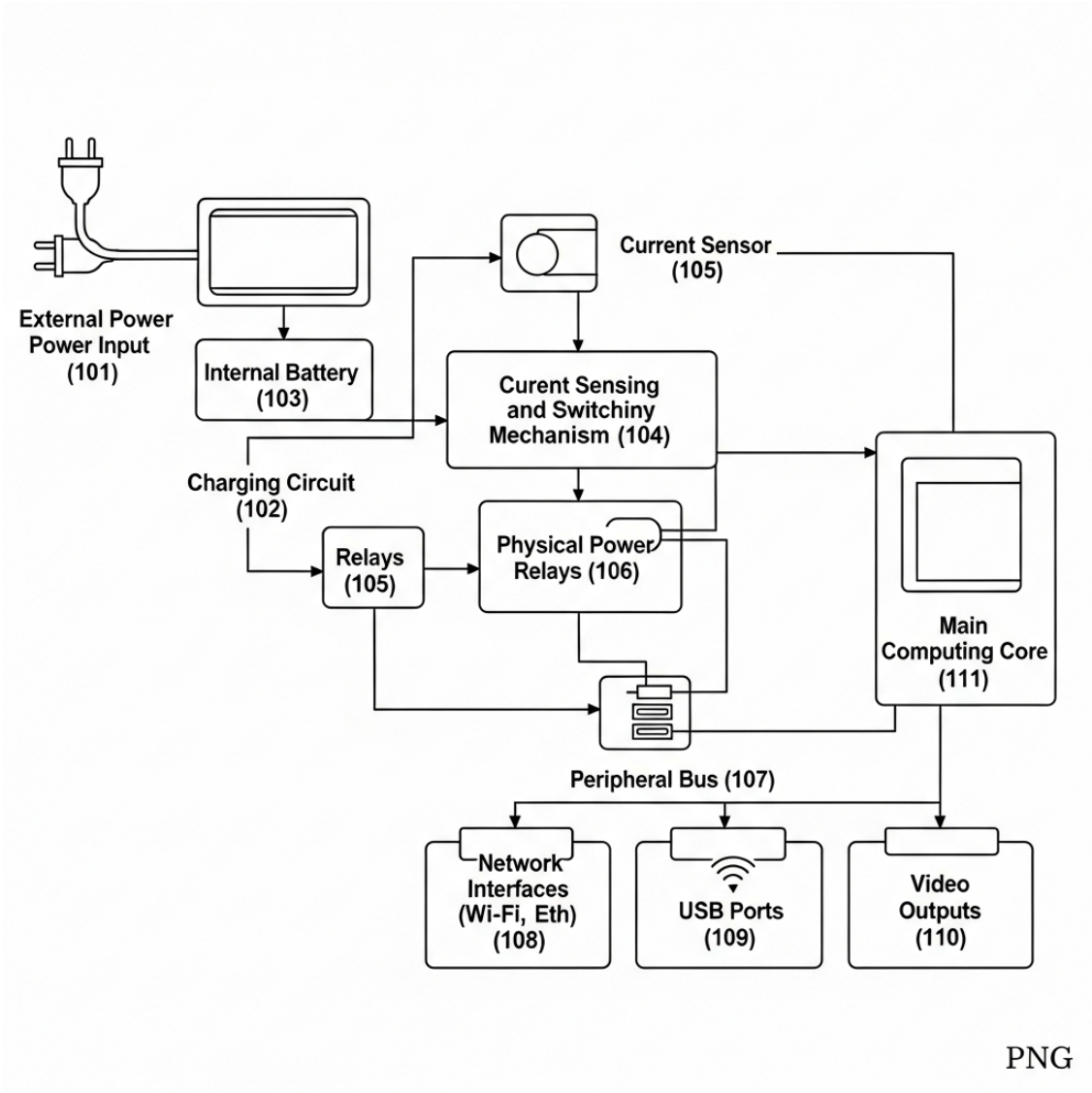


Figure 1: Block diagram of the secure computing device architecture.

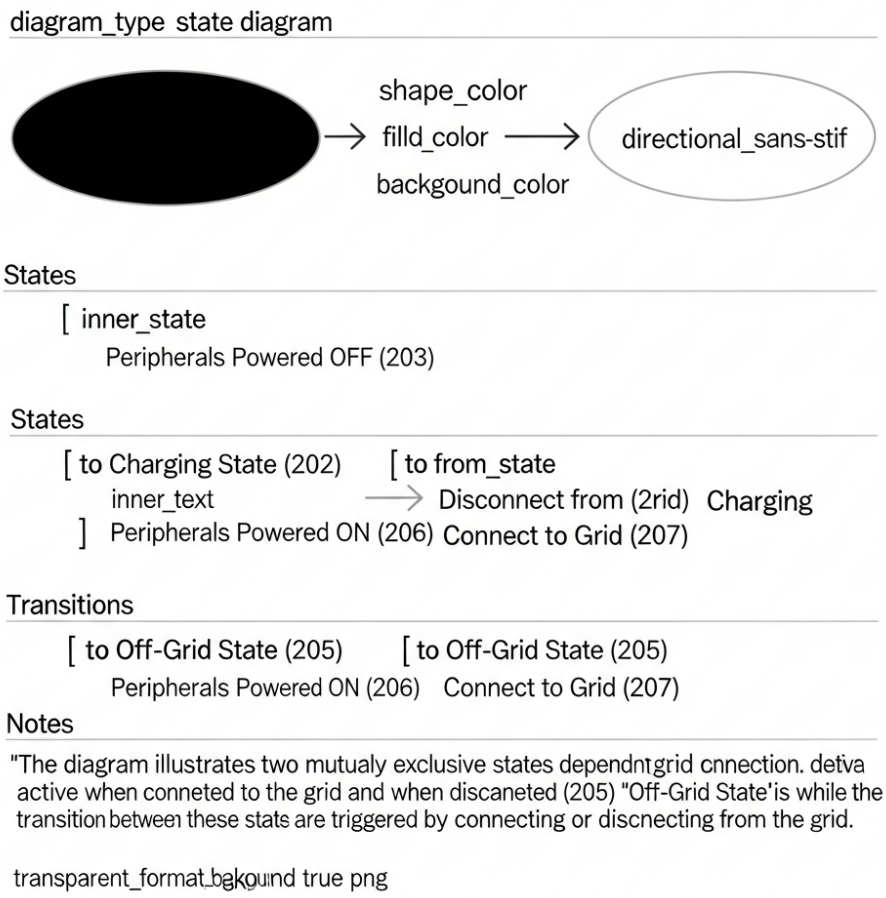


Figure 2: State diagram of operational states.